

PROJECT EXHIBITION I PRESENTATION

AI WOMEN SAFETY APP

Member-Wise Work Distribution & 5-Week Implementation Roadmap

Team Lead: **Vaishnavi Mishra (25BCE10943)**

Team Size: **5 Members**

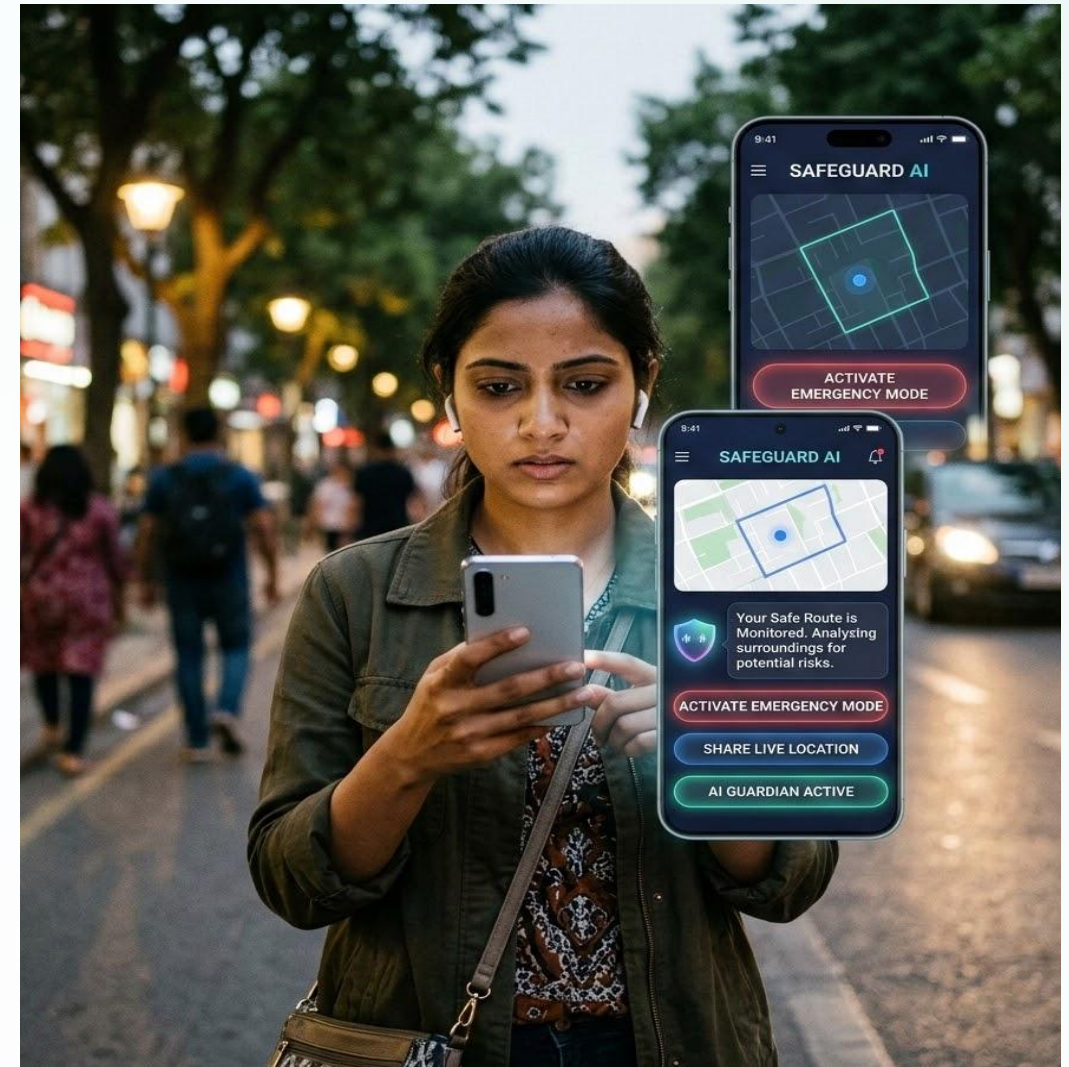
Timeline: **August - October 31**

Project Overview & Architecture

Explainable Emergency Solution

An intelligent, fast, and fully explainable safety application designed for women in distress, divided into 5 clear modules:

- **Responsive Frontend:** Modern UI built with React, Tailwind CSS, & Shadcn UI.
- **Auth & User Backend:** Secure authentication & emergency contacts database.
- **SOS Dispatcher:** Instant alert trigger sending emergency SMS via Twilio API.
- **GPS Location Module:** Real-time coordinate fetching & Google Maps URL generator.
- **Voice AI & Cloud Deploy:** Emergency phrase detection & live Firebase hosting.



Team Structure & Member Allocations

Member Name	Reg. Number	Designated Role	Core Module Responsibilities
Vaishnavi Mishra	25BCE10943	Team Leader (Frontend)	React UI Development, Component Styling, REST API Integration, UX Polish.
Princy Mahajan	25BCE10376	Backend (Auth & Users)	User Registration/Login APIs, Bcrypt Password Hashing, Contacts Database.
Aastha	25BCE10812	Backend (SOS Alert)	SOS Button Handler Route, Twilio SMS Dispatcher, Alert Notifications & Logs.
Pragati Sachdeva	25BCE10811	Backend (Live Location)	Browser Geolocation API, Real-time GPS Tracking, Google Maps Link Sharing.
Siya Julka	25BCE10477	Voice AI & Cloud Deploy	Emergency Voice Phrase Trigger, Speech Recognition AI, Firebase Cloud Hosting.

Technology Stack Breakdown



Frontend Layer

Vaishnavi Mishra

HTML5, CSS3, JavaScript, React / React Native, Tailwind CSS, Shadcn UI / Material UI, Axios API Client.



Backend & Security

Princy, Aastha & Pragati

Node.js & Express REST APIs, MongoDB Database, Bcrypt Password Hashing, Twilio SMS API, Geolocation API.



AI & Cloud Infra

Siya Julka

Web Speech API for AI phrase recognition ("help" / panic keywords), Firebase Hosting, and Cloud Infrastructure.

Vaishnavi Mishra — Weekly Plan

25BCE10943

👑 Team Leader (Frontend Developer)

WEEK 1

Setup & Design

- Set up React project skeleton & Git repository.
- Design Figma wireframes for Home, Login, SOS, & Contacts.
- Configure Tailwind CSS & color theme.

WEEK 2

Core UI Screens

- Build Home screen & main Navigation Bar.
- Develop Login & Register form UI components.
- Configure Axios API client instance.

WEEK 3

API Connectivity

- Build Emergency Contacts management screen UI.
- Connect Auth forms to Princy's backend APIs.
- Add form input validation & state feedback.

WEEK 4

Polish & Responsive

- Add loading states & error toast messages.
- Test mobile & tablet screen responsiveness.
- Refine UI consistency across all screens.

WEEK 5

Walkthrough & Viva

- Fix UI/UX bugs identified during testing.
- Prepare short frontend UI walkthrough demo.
- Document React state management code.

Princy Mahajan — Weekly Plan

25BCE10376

 Backend Developer (Auth & User Database)

WEEK 1

Database Architecture

- Design MongoDB schemas for Users & Contacts.
- Initialize Express.js server boilerplate.
- Document Auth REST API endpoints for team.

WEEK 2

Auth API Development

- Implement User Registration API route.
- Build User Login API route with checks.
- Develop Add & View Emergency Contacts APIs.

WEEK 3

Password Hashing

- Integrate Bcrypt for secure password hashing.
- Add email format & empty field checks.
- Test all Auth routes in Postman.

WEEK 4

Integration & Audit

- Integrate Auth endpoints with React frontend.
- Perform security audit on stored credentials.
- Handle duplicate user & bad token edge cases.

WEEK 5

Code Defense

- Resolve database connection & query bugs.
- Add inline comments for Express routes.
- Prepare explanations on password hashing.

Aastha — Weekly Plan

25BCE10812

🔔 Backend Developer (SOS Alert & SMS System)

WEEK 1

SMS Gateway Setup

- Setup Twilio trial developer account.
- Define SOS payload (User, Coordinates, Link).
- Map out emergency dispatch execution flow.

WEEK 2

SOS Handler Route

- Build backend route receiving SOS trigger.
- Connect Twilio API to send test SMS message.
- Create SOS Button component trigger handler.

WEEK 3

Broadcast Dispatch

- Fetch saved contacts from DB dynamically.
- Dispatch emergency SMS alerts to all contacts.
- Connect frontend SOS button to backend route.

WEEK 4

SMS Delivery Testing

- Conduct live SMS dispatch tests to mobile numbers.
- Handle edge cases (no contacts saved notice).
- Log alert history with timestamp in database.

WEEK 5

Flow Documentation

- Fix SMS dispatch delays & gateway errors.
- Comment SOS button to SMS dispatch flow.
- Practice step-by-step SOS flow for viva.

Pragati Sachdeva — Weekly Plan

25BCE10811

📍 Backend Developer (Live GPS Location Tracking)

WEEK 1

GPS API Research

- Test browser Geolocation API accuracy.
- Define Google Maps link format (``maps.google.com``).
- Design Location History collection schema.

WEEK 2

Coordinate Fetching

- Build "Get My Location" GPS fetch component.
- Display latitude & longitude coordinates on UI.
- Save coordinates & timestamp to database.

WEEK 3

Maps Link & History

- Build Location History screen UI & endpoint.
- Convert saved lat/lng into clickable Google Maps link.
- Verify link opens exact location on Maps.

WEEK 4

Mobile GPS Testing

- Test live GPS location on mobile browsers.
- Handle permission denial with friendly alerts.
- Optimize location update refresh rates.

WEEK 5

Code Commenting

- Fix location accuracy & timeout bugs.
- Document Geolocation & Maps link generation code.
- Prepare viva defense on GPS vs SDK trade-offs.

Siya Julka — Weekly Plan

25BCE10477

🎤 AI Voice Activation & Cloud Deployment

WEEK 1

Speech API & Firebase

- Test Web Speech API keyword detection ("help").
- Setup Firebase project (Spark Plan) instance.
- Review Firebase Hosting deployment steps.

WEEK 2

Voice Trigger Feature

- Build "Start Listening" voice trigger component.
- Detect emergency phrase "help" on speech input.
- Deploy initial test build build to Firebase.

WEEK 3

Voice-to-SOS Integration

- Connect detected voice phrase to Aastha's SOS function.
- Add visual "Listening..." indicator on UI.
- Redeploy updated project build to Firebase.

WEEK 4

Acoustic Tuning & Deploy

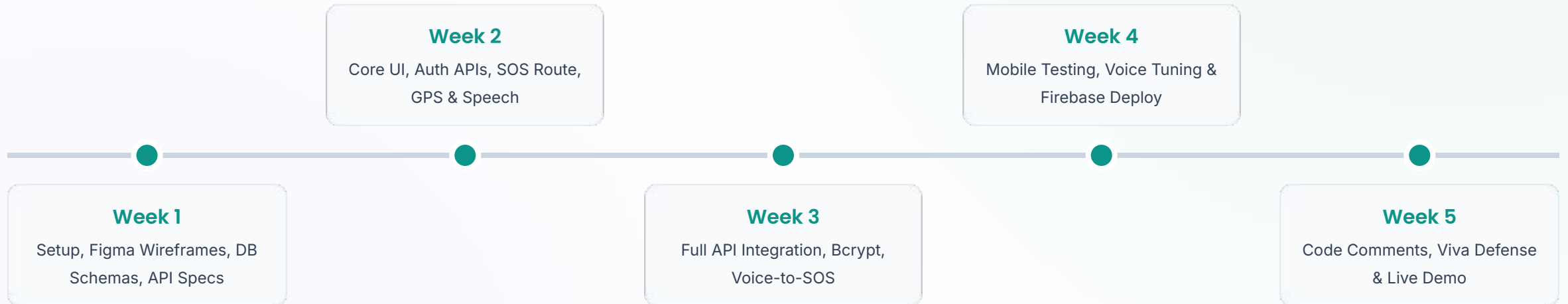
- Test voice trigger in quiet vs. noisy environments.
- Perform final full-app Firebase deployment.
- Verify live URL accessibility across devices.

WEEK 5

Live Demo Prep

- Fix speech recognition sensitivity & hosting bugs.
- Comment speech recognition & deployment scripts.
- Prepare live Firebase demo link for viva.

Master 5-Week Integration Roadmap



Thank You!

AI Women Safety App — Project Exhibition I Presentation